Week-2

Basic Statistical Functions for Data Exploration:

1. Measures of central tendency – mean, median, mode
2. Measures of data spread
3. Dispersion of data – variance, standard deviation
4. Position of the different data values – quartiles, inter-quartile range (IQR)

```
# importin all libraries
import pandas as pd
import numpy as np

#reading the data
df = pd.read_csv("mpg_data.csv")
df
```

```
      mpg  cylinders  displacement horsepower  weight  acceleration  \
0    15.0          8         350.0        165    3693          11.5
1    18.0          8         318.0        150    3436          11.0
2    16.0          8         304.0        150    3433          12.0
3    17.0          8         302.0        140    3449          10.5
4    15.0          8         429.0        198    4341          10.0
5    14.0          8         454.0        220    4354           9.0
6    14.0          8         440.0        215    4312           8.5
7    14.0          8         455.0        225    4425          10.0
8    15.0          8         390.0        190    3850           8.5
9    15.0          8         383.0        170    3563          10.0
10   14.0          8         340.0        160    3609           8.0
11   15.0          8         400.0        150    3761           9.5
12   14.0          8         455.0        225    3086          10.0
13   24.0          4         113.0         95    2372          15.0
14   22.0          6         198.0         95    2833          15.5
15   18.0          6         199.0         97    2774          15.5
16   21.0          6         200.0         85    2587          16.0
17   27.0          4          97.0         88    2130          14.5
18   26.0          4          97.0         46    1835          20.5
19   25.0          4         110.0         87    2672          17.5
20   24.0          4         107.0         90    2430          14.5
21   25.0          4         104.0         95    2375          17.5
22   26.0          4         121.0        113    2234          12.5
23   21.0          6         199.0         90    2648          15.0
24   10.0          8         360.0        215    4615          14.0
25   10.0          8         307.0        200    4376          15.0
26   11.0          8         318.0        210    4382          13.5
```

27	9.0	8	304.0	193	4732	18.5
28	27.0	4	97.0	88	2130	14.5
29	28.0	4	140.0	90	2264	15.5
..	...	...	...	...	...	...
363	28.0	4	112.0	88	2605	19.6
364	27.0	4	112.0	88	2640	18.6
365	34.0	4	112.0	88	2395	18.0
366	31.0	4	112.0	85	2575	16.2
367	29.0	4	135.0	84	2525	16.0
368	27.0	4	151.0	90	2735	18.0
369	24.0	4	140.0	92	2865	16.4
370	36.0	4	105.0	74	1980	15.3
371	37.0	4	91.0	68	2025	18.2
372	31.0	4	91.0	68	1970	17.6
373	38.0	4	105.0	63	2125	14.7
374	36.0	4	98.0	70	2125	17.3
375	36.0	4	120.0	88	2160	14.5
376	36.0	4	107.0	75	2205	14.5
377	34.0	4	108.0	70	2245	16.9
378	38.0	4	91.0	67	1965	15.0
379	32.0	4	91.0	67	1965	15.7
380	38.0	4	91.0	67	1995	16.2
381	25.0	6	181.0	110	2945	16.4
382	38.0	6	262.0	85	3015	17.0
383	26.0	4	156.0	92	2585	14.5
384	22.0	6	232.0	112	2835	14.7
385	32.0	4	144.0	96	2665	13.9
386	36.0	4	135.0	84	2370	13.0
387	27.0	4	151.0	90	2950	17.3
388	27.0	4	140.0	86	2790	15.6
389	44.0	4	97.0	52	2130	24.6
390	32.0	4	135.0	84	2295	11.6
391	28.0	4	120.0	79	2625	18.6
392	31.0	4	119.0	82	2720	19.4

	model year	origin	car name
0	70	1	"buick skylark 320"
1	70	1	"plymouth satellite"
2	70	1	"amc rebel sst"
3	70	1	"ford torino"
4	70	1	"ford galaxie 500"
5	70	1	"chevrolet impala"
6	70	1	"plymouth fury iii"
7	70	1	"pontiac catalina"
8	70	1	"amc ambassador dpl"
9	70	1	"dodge challenger se"
10	70	1	"plymouth 'cuda 340"
11	70	1	"chevrolet monte carlo"
12	70	1	"buick estate wagon (sw)"
13	70	3	"toyota corona mark ii"
14	70	1	"plymouth duster"
15	70	1	"amc hornet"
16	70	1	"ford maverick"
17	70	3	"datsun pl510"
18	70	2	"volkswagen 1131 deluxe sedan"
19	70	2	"peugeot 504"

```
20          70       2                        "audi 100 ls"
21          70       2                           "saab 99e"
22          70       2                           "bmw 2002"
23          70       1                        "amc gremlin"
24          70       1                          "ford f250"
25          70       1                          "chevy c20"
26          70       1                         "dodge d200"
27          70       1                           "hi 1200d"
28          71       3                       "datsun pl510"
29          71       1                 "chevrolet vega 2300"
..         ...     ...                                   ...
363         82       1                  "chevrolet cavalier"
364         82       1            "chevrolet cavalier wagon"
365         82       1           "chevrolet cavalier 2-door"
366         82       1          "pontiac j2000 se hatchback"
367         82       1                      "dodge aries se"
368         82       1                     "pontiac phoenix"
369         82       1                "ford fairmont futura"
370         82       2                 "volkswagen rabbit l"
371         82       3                  "mazda glc custom l"
372         82       3                    "mazda glc custom"
373         82       1              "plymouth horizon miser"
374         82       1                      "mercury lynx l"
375         82       3                    "nissan stanza xe"
376         82       3                        "honda accord"
377         82       3                      "toyota corolla"
378         82       3                          "honda civic"
379         82       3                  "honda civic (auto)"
380         82       3                       "datsun 310 gx"
381         82       1               "buick century limited"
382         82       1  "oldsmobile cutlass ciera (diesel)"
383         82       1          "chrysler lebaron medallion"
384         82       1                      "ford granada l"
385         82       3                    "toyota celica gt"
386         82       1                   "dodge charger 2.2"
387         82       1                    "chevrolet camaro"
388         82       1                     "ford mustang gl"
389         82       2                           "vw pickup"
390         82       1                       "dodge rampage"
391         82       1                         "ford ranger"
392         82       1                          "chevy s-10"

[393 rows x 9 columns]
```

1. Measure of contral tendency - Mean, Median, Mode

```
#Mean
df['displacement'].mean()
```

```
193.89440203562341
```

```
#Median
df['displacement'].median()
```

```
151.0
```

```
#Mode
df['displacement'].mode()
```

```
0    97.0
dtype: float64
```

```
#count of each value in the col
df['displacement'].value_counts()
```

```
97.0     21
350.0    18
98.0     18
318.0    17
250.0    17
140.0    15
225.0    13
400.0    13
91.0     12
232.0    11
121.0    11
302.0    11
151.0     9
120.0     9
200.0     8
351.0     8
90.0      8
231.0     8
304.0     7
122.0     7
105.0     7
85.0      7
79.0      6
119.0     6
156.0     6
89.0      5
258.0     5
107.0     5
135.0     5
108.0     5
         ..
262.0     2
199.0     2
155.0     1
110.0     1
181.0     1
144.0     1
340.0     1
101.0     1
130.0     1
104.0     1
390.0     1
454.0     1
115.0     1
80.0      1
171.0     1
68.0      1
145.0     1
```

```
81.0      1
78.0      1
131.0     1
183.0     1
83.0      1
111.0     1
88.0      1
72.0      1
76.0      1
97.5      1
114.0     1
96.0      1
267.0     1
Name: displacement, Length: 81, dtype: int64
```

2. Measures of data spread

```
df.describe()
```

```
              mpg   cylinders  displacement       weight  acceleration  \
count  393.000000  393.000000    393.000000   393.000000    393.000000
mean    23.457506    5.463104    193.894402  2973.613232     15.562850
std      7.791557    1.700626    104.468271   849.212695      2.755976
min      9.000000    3.000000     68.000000  1613.000000      8.000000
25%     17.000000    4.000000    105.000000  2223.000000     13.800000
50%     23.000000    4.000000    151.000000  2800.000000     15.500000
75%     29.000000    8.000000    262.000000  3613.000000     17.100000
max     46.600000    8.000000    455.000000  5140.000000     24.800000

       model year      origin
count  393.000000  393.000000
mean    75.977099    1.575064
std      3.676592    0.805016
min     70.000000    1.000000
25%     73.000000    1.000000
50%     76.000000    1.000000
75%     79.000000    2.000000
max     82.000000    3.000000
```

3. Dispersion of data – variance, standard deviation

```
# standard deviation for each column
df.std()
```

```
mpg               7.791557
cylinders         1.700626
displacement    104.468271
weight          849.212695
acceleration      2.755976
model year        3.676592
origin            0.805016
dtype: float64
```

```
#Varaince of all columns
columns = ['mpg', 'cylinders', 'displacement']
```

```
print("Print variance for each Column")
# for col in columns:
#     print(df[col].var())
df.var()
```

```
Print variance for each Column

mpg                 60.708368
cylinders            2.892130
displacement     10913.619560
weight          721162.202069
acceleration         7.595402
model year          13.517331
origin               0.648050
dtype: float64
```

4. **Position of the different data values – quartiles, inter-quartile range (IQR).**

```
q1 = np.percentile(df['displacement'], 25, interpolation='midpoint')
q1
```

```
105.0
```

```
q2 = np.percentile(df['displacement'], 50, interpolation='midpoint')
q2
```

```
151.0
```

```
q3 = np.percentile(df['displacement'], 75, interpolation='midpoint')
q3
```

```
262.0
```

```
# Inter quartile Range
IQR = q3-q1
IQR
```

```
157.0
```

```
df['displacement'].quantile(0.25)
```

```
105.0
```